

# Title = Civics & Community Engagement

Roslan

## Subjective

### **Write short answers: (10)**

- ✓ 1- Define digital divide.
- 2- How to improve public discourse in Pakistan?
- ✓ 3- Explain the importance of public opinion.
- ✓ 4- What is diversity?
- 5- Write two advantages of digital platform.

### **NOTE: Attempt all "Long Questions"**

- 1- Write a detailed note on Agha Khan Rural Development Program. (10)
- 2- Define the term 'Culture' and its characteristics. Elaborate how culture shapes our thoughts, behaviors, and personality. (10)

# Lahore College for Women University, Lahore

## DATABASE SYSTEMS

### MID TERM EXAMINATION

Q # 1

#### Case Study:

A University contains many Faculties. The Faculties in turn are divided into several Schools. Each School offers numerous programs and each program contains many courses. Lecturers can teach many different courses and even the same course numerous times. Courses can also be taught by many lecturers. A student is enrolled in only one program but a program can contain many students. Students can be enrolled in many courses at the same time and the courses have many students enrolled. Draw an ER Diagram for the above scenario.

(Marks: 5/10)

Q # 2

a) Describe three level of ANSI/ SPARC/ 3-Tier model with diagram

(Marks: 5)

b) Explain normalization and discuss its first 3 forms.

(Marks: 5)

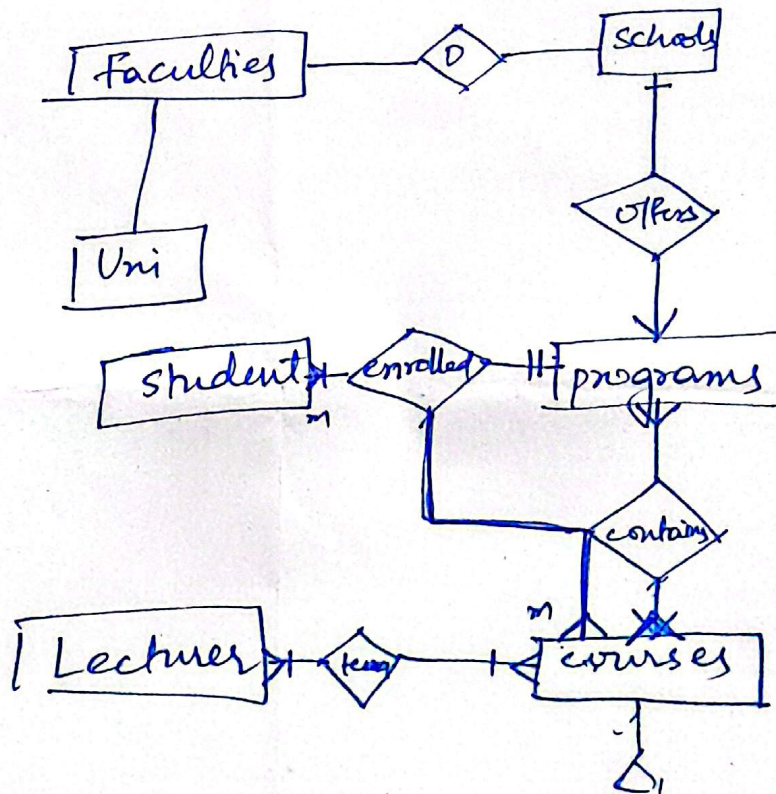
Q # 3 Describe the following: any (2)

(Marks: 10)

a) Unary, binary and ternary relationship

b) Strong and weak entities

c) Simple and composite attribute



**LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE**  
**Final Examination Spring 2025**

**Class: BSCS**

**Subject: Computer Science**

**Semester: IV**

**Term: Final**

**Paper Title: Operating Systems**

**Max Marks: 50**

**Time Allowed: 3 hours**

- Q:1** (a) Consider the following scenario: two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes: Here, wants1 and wants2 are shared variables, which are initialized to false. **[5+5=10]**

|                                                                                                                                                                                    |                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>/* P1 */ while (true) {     wants1 = true;     while (wants2 == true):         /* Critical         Section */         wants1=false;     }     /* Remainder section */ }</pre> | <pre>/* P2 */ while (true) {     wants2 = true;     while (wants1==true);         /* Critical         Section */         wants2 = false;     }     /* Remainder section */ }</pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Either the following statements are true or false about the above construct?

|                                                                              |  |
|------------------------------------------------------------------------------|--|
| It does not ensure mutual exclusion.                                         |  |
| It does not ensure bounded waiting.                                          |  |
| It requires that processes enter the critical section in strict alternation. |  |
| It does not prevent deadlocks, but ensures mutual exclusion.                 |  |

- (b) Explain the Security violation categories.

- Q:2** Consider the following scenarios and answer the questions:

**[5+5=10]**

- (a) Resource Allocation Graphs can be used to detect deadlocks from a system. Evaluate the following system:

**Processes:** P1, P2, P3

**Resources:** R1, R2, R3

**Edges:** P1 is waiting for a resource R1

P2 is waiting for a resource R2

P3 is waiting for a resource R3

R1 is allocated to P2

R2 is allocated to P3

R3 is allocated to P1

Is there a deadlock in this system? Justify your answer

3

*[Handwritten signature]*

P.T.O  
*[Handwritten signature]*

(b) A single processor system has three resource types X, Y and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column alloc denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution. Write the sequence of process execution?

|    | Alloc |   |   | Request |   |   |
|----|-------|---|---|---------|---|---|
|    | X     | Y | Z | X       | Y | Z |
| P0 | 1     | 2 | 1 | 1       | 0 | 3 |
| P1 | 2     | 0 | 1 | 0       | 1 | 2 |
| P2 | 2     | 2 | 1 | 1       | 2 | 0 |
|    | 0     | 1 | 2 |         |   |   |

Q:3 Consider a disk with cylinders numbered from 0 to 199. The disk queue contains the following set of requests in the order they arrive: [10]

Disk Queue: 180, 40, 20, 75, 90, 150, 110, 130  
The initial position of the disk head is at cylinder 120, and the head is currently moving toward higher-numbered cylinders.

- Determine the order in which the requests will be serviced using the following disk scheduling algorithms:
  - SCAN (Elevator Algorithm)
  - C-SCAN (Circular SCAN)
  - C-LOOK (Circular LOOK)
- For each algorithm, show the head movement and calculate the **total head movements** in terms of cylinders.

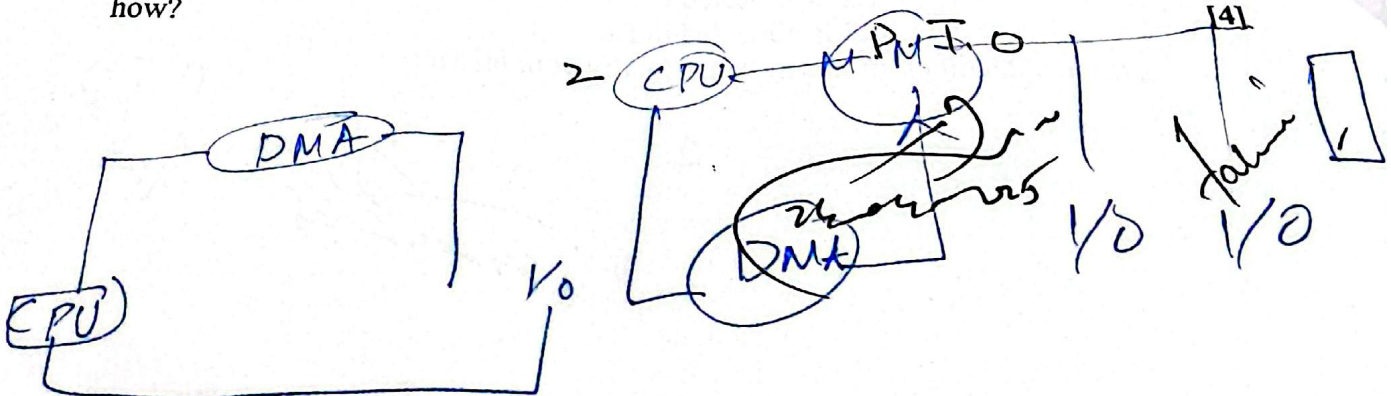
Q4: (a) Compare the Algorithms

Page Reference String: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7

Number of Page Frames: 4

Compute the number of page faults for the following algorithms: FIFO, LRU and Optimal and elaborate which algorithm performs best for this reference string? [6]

(b) Define how Direct Memory Access helps the hardware to access main memory. Discuss how?



**LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE**

**FINAL TERM EXAMINATION – 2025**

**095**

**CLASS: BS- II**

**SUBJECT: GEDU-239- PAKISTAN STUDIES**

**SEMESTER: IV**

**CREDIT HOURS: 2**

**MAX MARKS: 50**

**TIME ALLOWED: 1 Hour & 40 Min**

**Note: Attempt all questions**

**SUBJECTIVE (30 Marks)**

2. Give short answers.

(10)

- i. What is the impact of climate change on Pakistan's agriculture.
- ii. Narrate any two features of Indus Valley Civilization.
- iii. Write about the importance of Northern Mountains.
- iv. How can Pakistan improve its economic growth rate.
- v. The name of Pakistan was suggested by whom and when.

3. Discuss the role of Quaid e Azam Muhammad Ali Jinnah as the first Governor General of Pakistan.

(10)

4. Write a detailed note on Pakistan India relationships.

(10)

*Dr. Sohail Nadeem*

Set-2

**LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE**

**Final Term Examinations, 2025**

*Rizhan*

**Paper Title: Entrepreneurship Code: GEDU-251 Session: 2023 -2027**

**Semester: IV**

**Max Marks: 50**

**Total Time: 3 hours**

**SUBJECTIVE SECTION**

**Attempt all questions**

**Question No1: Short Questions**

**(4\*5=20)**

- ✓1. Explain the concept of "window of opportunity" with an example.
- ✓2. Why is organizational feasibility important for a new business?
- ✓3. Mention the differences between content writing and SEO in digital marketing.
4. What legal compliances are essential for starting a business?

**Question No 2: Discuss how new venture can ensure data privacy and protect their intellectual property.**

**(15)**

**Question No 3: Suggest ways how businesses can use social media marketing to build their brand visibility.**

**(15)**

*H*

*C.O (196)  
T.M (200)  
P. (200)*

**Lahore College for Women University**  
**Department of Computer Science**  
**Final Term Exam Spring 2025**

**Professional Practices**  
**Duration: 3 Hours**

**Program: BSCS (Semester 4)**

**Session: 2023-2027**  
**Total Marks: 50**

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**Subjective Section**

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**Question 3: Short Questions**

**(15 Marks)**

1. List five functions of a software house.
  2. Mention five professional ethics every IT professional should follow.
  3. List five types of software misuse.
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**Question 4: Software License Violation in a Startup**

**(15 Marks)**

You are working as a junior developer at a tech startup. One day, you discover that your team is using a premium software tool without a valid license to speed up development. When you raise your concern with your manager, you are told to "just ignore it" because "everyone does it and we're a small company."

**Question:**

As a professional in the IT field, what actions would you take in this situation? Justify your response with reference to professional ethics, copyright law, and software licensing.

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*Saima*  
*10-04-2025*

**SUBJECT: Civics and Community Engagement**  
**MID TERM EXAM**  
**Department of Computer Science**  
**LAHORE COLLEGE FOR WOMEN UNIVERSITY**

**TOTAL MARKS-30**

**TIME DURATION- 2H**

**Short Questions (5 marks for each question)**

1. Explain the difference between Civics and citizenship?
2. What is the importance of community engagement?

**Long Questions**

**Note: Each question carry 10 marks. Attempt any two questions**

3. What is ethical consideration in civics engagement? Explain in detail ✓
4. What's the history of civics participation in the digital age and globalization (1990-present)
5. Write a detail note on the fundamental rights according to 1973 constitution?